

Decoding Currency Dynamics: A Hybrid CNN–LSTM–Transformer with Multi-Modal Fusion and Honest Regime-Aware Evaluation for Multi-Step Gold Price Forecasting

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Abstract—Foreign-exchange forecasting is difficult because prices are driven simultaneously by price action, macroeconomic fundamentals and market sentiment, while classical models such as ARIMA and GARCH assume the linearity and stationarity that currency data routinely violate. We present Decoding Currency Dynamics, an AI-driven framework for multi-step forecasting of the XAU/USD (gold) exchange rate. Its core is a dual-tower Hybrid CNN–LSTM–Transformer (≈ 4.39 M parameters) that fuses three real data streams—twelve technical indicators, six stationary macroeconomic variables, and thirteen FinBERT news-sentiment features—and predicts the next ten daily log-returns together with a calibrated uncertainty band. A dilated causal CNN extracts local structure, a presence-gated cross-attention module fuses sparse news into the price representation, a Transformer supplies global context ahead of parallel Bi-LSTM/Bi-GRU recurrence, and regime-conditioned probabilistic heads emit a mean and variance per horizon under a Gaussian negative-log-likelihood objective. Four training techniques—two-stage freeze-and-tune, probabilistic (GARCH-emulating) heads, modality masking and deep supervision—stabilise learning on sparse, heterogeneous data. Evaluated by strict walk-forward over 25.9 years of daily data (963 test origins, three seeds), the hybrid reaches a directional accuracy of 0.534 ± 0.001 with the lowest mean absolute error among deep configurations. We report results honestly: an AR(1)–GARCH(1,1) baseline still leads unfiltered directional accuracy (0.575), and a placebo-controlled ablation isolates a genuine but modest (≈ 1.4 percentage-point) contribution from a news-diffusion feature, of which the majority of the naive gain is shown to be an added-channel effect. The hybrid’s advantages are multi-modal reach, native uncertainty quantification and a conviction-based decision layer rather than raw accuracy.

Index Terms—foreign-exchange forecasting, deep learning, CNN, LSTM, Transformer, FinBERT sentiment, multi-modal fusion, GARCH, probabilistic forecasting, ablation study

I. Introduction

Foreign-exchange (FX) markets are among the most liquid and volatile financial markets, influenced by macroeconomic indicators, geopolitical events and market sentiment. Traditional forecasting approaches such as the Box–Jenkins ARIMA framework [1] and Bollerslev’s GARCH model [2] remain foundational for autocorrelation and

volatility clustering, but both presuppose stationarity and linear dependence structures that high-frequency currency data violate, motivating a shift toward non-linear, data-driven and increasingly multi-modal architectures.

Multi-step forecasting is especially hard: errors compound across horizons, and most prior deep-learning studies report only single-step accuracy. This paper targets the harder multi-horizon problem for gold (XAU/USD), a liquid and strongly macro-sensitive instrument. Our contributions are:

- 1) A dual-tower Hybrid CNN–LSTM–Transformer that unifies convolutional spatial extraction, recurrent temporal modelling, Transformer global attention and FinBERT sentiment fusion in a single end-to-end, regime-aware, probabilistic architecture.
- 2) A multi-modal feature-fusion pipeline that aligns technical, macroeconomic and news-sentiment streams onto a common daily grid, with a presence gate that keeps the model robust when news is absent (the common case).
- 3) Four training techniques—two-stage freeze-and-tune, Gaussian-NLL probabilistic heads, modality masking and deep supervision—that make training stable on sparse, heterogeneous data.
- 4) An honest, leakage-free evaluation protocol: full walk-forward against ARIMA and GARCH at every test origin, a placebo-controlled ablation, and explicit reporting of where the classical baseline still wins.

II. Related Work

Hochreiter and Schmidhuber’s Long Short-Term Memory [3] introduced the gating mechanism that underpins nearly every sequential financial forecaster; Vaswani et al. [4] showed that self-attention captures long-range dependencies without recurrence. Recent IEEE-indexed work converges on themes directly relevant here. Benchmarks of LSTM, gradient-boosted trees and Transformers

on multiple currency pairs [7] conclude that no single architecture dominates all regimes and that an ensemble balances accuracy and reliability, motivating a hybrid rather than single-model design. Sequential deep-learning studies that fuse windowed LSTM encoders with news sentiment [8] outperform ARIMA baselines, reinforcing the value of pairing sequence models with exogenous text. On the spatial front, converting price series into image-like representations for a convolutional classifier [9] achieves high directional accuracy on out-of-sample data, justifying a CNN local-feature stage. Most directly, augmenting an econometric FX model with a FinBERT [5] sentiment index [10] shows the index behaves as a leading indicator—the principal precedent for our sentiment backbone. Three gaps persist: these streams are usually treated as separate pipelines rather than one unified model; multi-step forecasting is rarely addressed; and evaluation seldom isolates behaviour by volatility regime. Our design targets all three.

III. Data and Feature Engineering

A. Data sources

Daily XAU/USD candles are obtained from a public market feed (ticker GC=F), giving 6,487 bars spanning 2000–2026 (≈ 25.9 years). Macroeconomic variables are sourced from public providers—short-rate and Treasury-yield proxies, the trade-weighted dollar index, and Consumer-Price-Index releases. Financial news headlines are collected from a free news API and scored for sentiment. The pipeline is incremental and cached: news is fetched only since the last run and each headline is FinBERT-scored once, so historical text is never re-scored.

B. Three aligned streams (31 features)

All streams are computed over an identical 60-bar lookback and fused into a per-bar vector. Technical (12): OHLC log-returns, RSI, MACD histogram, Bollinger-band width, volume z -score, ATR%, rate-of-change, stochastic %K and an EMA ratio. Macro (6): short-rate z -score, 10-year yield change, dollar-index return, CPI year-over-year and month-over-month, and days-since-CPI. Sentiment (13): nine rolling FinBERT statistics—including a diffusion breadth index—plus a one-hot buy/sell/hold/none trading signal. All features use stationary transforms and are normalised with statistics fit on the training split only.

C. FinBERT sentiment scoring

FinBERT [5], a BERT-family transformer pretrained on financial text, emits a three-way distribution over {positive, neutral, negative} per headline. Following industry practice, we aggregate only directional headlines ($|\text{polarity}| \geq 0.15$, confidence-weighted) so that the neutral majority does not dilute the signal, and we derive the buy/sell trading signal with an adaptive, regime-relative threshold (an expanding, causal z -score) rather than a fixed cut. A diffusion breadth index ($\# \text{bullish} -$

Quant ($B, 60, 18$) $\rightarrow$ Dilated CNN ($B, 60, 128$)
 Text ($B, 60, 13$) $\rightarrow$ GRU ($B, 60, 128$)
 $\Downarrow$ Cross-Attention (+presence gate) ($B, 60, 128$)
 $\Downarrow$ Transformer ($B, 60, 256$)
 $\Downarrow$ Bi-LSTM $\parallel$ Bi-GRU + pool ($B, 256$)
 $\Downarrow$ Regime heads (μ, σ^2) $\times 10 \oplus$ XGBoost expert

Fig. 1. Dual-tower Hybrid CNN–LSTM–Transformer data flow with tensor shapes. (Replace with the exported architecture figure for camera-ready.)

$\# \text{bearish} / \# \text{total}$ captures directional agreement independently of magnitude.

IV. Model Architecture

The network (Fig. 1) is a dual-tower design. Tensor shapes are given for batch size B and lookback $T = 60$.

A. Tower A—quantitative pathway

The 18 technical+macro features ($B, 60, 18$) are projected to width 64 and passed to a dilated causal CNN: three stacked convolution blocks with dilations 1/2/4 and left-padding (WaveNet/TCN style), no pooling, preserving full 60-bar resolution $\rightarrow (B, 60, 128)$. A learned volatility-regime embedding is added.

B. Tower B—news pathway

The 13 sentiment features ($B, 60, 13$) are encoded by a GRU and projected to ($B, 60, 128$).

C. Presence-gated cross-attention fusion

Multi-head cross-attention uses the price stream as Query and the news stream as Key/Value. A per-timestep presence gate $g_t = \sigma(W_g \mathbf{s}_t)$ scales the news contribution on a residual path, so on news-less bars the gate closes and the model falls back to price/macro:

$$\mathbf{f}_t = \text{LN}(\mathbf{q}_t + g_t \cdot \text{Attn}(\mathbf{q}_t, \mathbf{K}, \mathbf{V})). \quad (1)$$

D. Transformer, recurrence and pooling

The fused sequence is projected to $d_{\text{model}} = 256$ and passed through a Transformer encoder with sinusoidal positional encoding and a causal mask $\rightarrow (B, 60, 256)$; global attention is placed before recurrence to avoid recency bias. Parallel Bi-LSTM $\parallel$ Bi-GRU branches are blended by a learned gate and attention-pooled to a context vector ($B, 256$).

E. Regime-aware probabilistic output and fused expert

A volatility-regime detector routes the context to regime-conditioned decoder heads that emit a mean μ_h and log-variance $\log \sigma_h^2$ for each of the ten horizons. A frozen XGBoost [6] expert is fused through a regime-driven per-horizon trust gate:

$$\hat{y}_h = \tau_h \odot \hat{y}_h^{\text{xgb}} + (1 - \tau_h) \odot \hat{y}_h^{\text{deep}}. \quad (2)$$

V. Training Methodology

A. Two-stage freeze-and-tune

Because news exists for only a fraction of history, training the text tower end-to-end would project many news-less years of zeroed sentiment onto the fusion. Stage 1 trains the quantitative backbone text-free over the full history; Stage 2 freezes that backbone (≈ 4.15 M parameters) and fine-tunes only the text tower, fusion and heads (≈ 0.24 M parameters) on the news-dense recent subset at $0.1\times$ the learning rate.

B. Gaussian NLL (probabilistic) heads

Each head predicts a distribution, trained under

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2} \left(\log \sigma^2 + \frac{(y - \mu)^2}{\sigma^2} \right), \quad (3)$$

emulating GARCH’s conditional-variance modelling inside the network. The predicted σ yields an uncertainty band and a conviction statistic $|\mu|/\sigma$ that powers a calibrated abstention rule and a costed backtest.

C. Modality masking and deep supervision

The sentiment stream is randomly zeroed with probability $p = 0.4$ during training so the model cannot over-rely on frequently-absent news; this complements the inference-time presence gate. An auxiliary head on a deeper expert (weight 0.5) injects gradient signal into the middle of the deep stack, improving flow and stabilising training. A robust Huber term and a small directional loss complete the objective.

VI. Experimental Setup

Data are split chronologically (70/15/15). Normalisation statistics are fit on the training split only. Both classical baselines—ARIMA (walk-forward) and AR(1)–GARCH(1,1)—are refit at every one of the 963 test origins the hybrid is scored on. We report the mean $\pm$ standard deviation across three seeds (9, 36, 99). Primary metrics are directional accuracy (sign agreement across all horizons), mean absolute error (MAE) and RMSE on cumulative log-returns; we additionally report a placebo-controlled ablation, a seed ensemble and a costed conviction backtest.

VII. Results and Discussion

A. Baseline comparison

Table I reports the walk-forward comparison. The AR(1)–GARCH(1,1) baseline leads unfiltered directional accuracy; the hybrid narrows the gap with near-zero seed variance and achieves the lowest MAE among deep configurations.

TABLE I
Walk-forward comparison (963 test origins, 3 seeds).

Model	Dir. Acc.↑	MAE↓	RMSE↓
GARCH (AR(1)–GARCH(1,1))	0.575	0.0196	0.0275
ARIMA (walk-forward)	0.485	0.0199	0.0278
Hybrid CNN–LSTM–Transformer	0.534 $\pm$ 0.001	0.0221	0.0304

TABLE II
Ablation of the sentiment diffusion feature (same seeds).

Configuration	Dir. Acc.
Without diffusion (30 features)	0.5006
Placebo—shuffled diffusion (31)	0.5209
With real diffusion (31 features)	0.5345

B. Ablation with placebo control

Table II isolates the sentiment diffusion feature using a same-seed A/B plus a placebo: replacing the feature with a random permutation of its own values (same distribution, signal destroyed). The naive +3.4 pp gain decomposes into ≈ 2.0 pp of added-input-channel effect (achieved even by noise) and ≈ 1.4 pp of genuine diffusion signal—a small but real, seed-stable contribution.

C. Conviction machinery

The seed ensemble reaches 0.534 directional accuracy overall and 0.549 at 10% coverage (its most-confident calls). The probabilistic conviction signal drives a costed 1-step strategy that returns +61% net at an annualised Sharpe of 0.76 over the test window—evidence that the model’s uncertainty is informative even where its unfiltered accuracy is not.

D. Honest discussion

On this strongly trending, volatility-clustered daily series, GARCH’s AR(1) conditional-mean drift is a genuinely strong directional baseline that the hybrid does not overtake on raw accuracy. A feature-attribution analysis attributes the forecast largely to technical/price features (≈ 70 –82%) and macro (≈ 18 –30%), with news sentiment contributing little—consistent with the sparse news coverage (≈ 14 –18% of bars). The hybrid’s value is therefore its breadth and decision quality: multi-modal inputs, learned non-linear structure, native per-horizon uncertainty, regime awareness, and a conviction layer that turns forecasts into a positive costed strategy.

VIII. Limitations

News coverage is the binding constraint: with only ≈ 14 –18% of test bars carrying headlines, the sentiment stream cannot express its full value, and denser coverage via a keyed news provider is the primary lever for improvement. Results are for a single instrument (XAU/USD) at daily cadence. Finally, we emphasise that directional accuracy

above ≈ 0.6 is not attainable on this task without look-ahead leakage; we report our figures as such rather than pursuing inflated numbers.

IX. Conclusion and Future Work

We presented a reproducible, multi-modal Hybrid CNN-LSTM-Transformer that unifies convolutional, recurrent, attention and sentiment pathways into one regime-aware, probabilistic forecaster, and evaluated it honestly against classical baselines with a placebo-controlled ablation. The framework closes the three literature gaps—unified multi-modal design, explicit multi-step forecasting, and regime-aware evaluation—while being transparent about where a strong econometric baseline still wins. Future work will (i) integrate a keyed, denser news source to unlock the sentiment stream, (ii) add explainable-AI attributions (SHAP, attention analysis), (iii) conduct broader ablations and multi-pair generalisation, and (iv) wrap the probabilistic forecaster in a reinforcement- learning execution agent.

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